

EEE4114F Class Test 2

May 2023

Duration: 1 hours

Total: 40 marks

Empl ID: _____

Instructions:

- This is a closed-book test but you may use a calculator.
- There are two parts to this test.
- **Part A** has *two* questions totalling 20 marks. You must answer all of them.
- **Part B** has *two* questions totalling 20 marks. You must answer all of them.
- Part A and part B each count 50% of the final mark.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs appears at the end of this paper.
- You have 1 hour.

Part A:	Question:	1	2	Total
	Points:	10	10	20
	Score:			

Part B:	Question:	3	4	Total
	Points:	10	10	20
	Score:			

Part A: Digital Signal Processing

Empl ID: _____

Question 1 [10 marks]

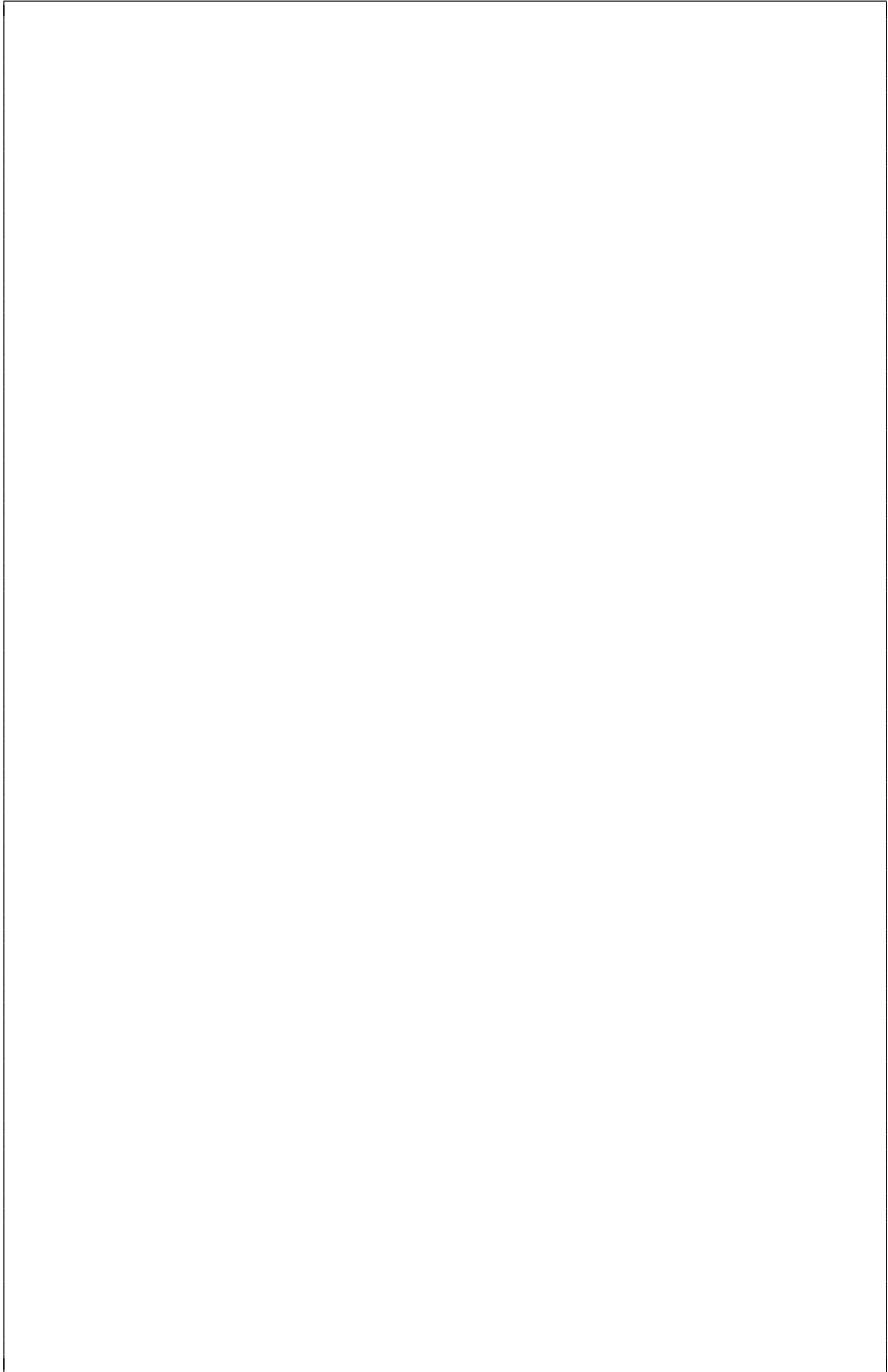
The impulse response of a discrete-time LTI system is

$$h[n] = -\delta[n] + 2 \left(\frac{1}{2} \right)^n u[n].$$

- Plot $h[n]$ over the range $n = -5$ to $n = 5$.
- Is the system causal? Why?
- Draw a pole-zero plot of the system function $H(z)$.
- Assuming that

$$H(z) = \frac{1 + 1/2z^{-1}}{1 - 1/2z^{-1}} \quad \text{for } |z| > 1/2,$$

or otherwise, find the impulse response $g[n]$ of the stable inverse of the system.



Question 2 [10 marks]

- a. The continuous-time signal $x_c(t) = \cos(40\pi t)$ is sampled with a sampling period T to obtain the discrete-time signal $x[n] = \cos(\pi n/15)$. Determine a choice for T consistent with this information.
- b. Is your choice for T above unique? If so, explain why. If not, specify another choice of T consistent with the information given.
- c. The continuous-time signal

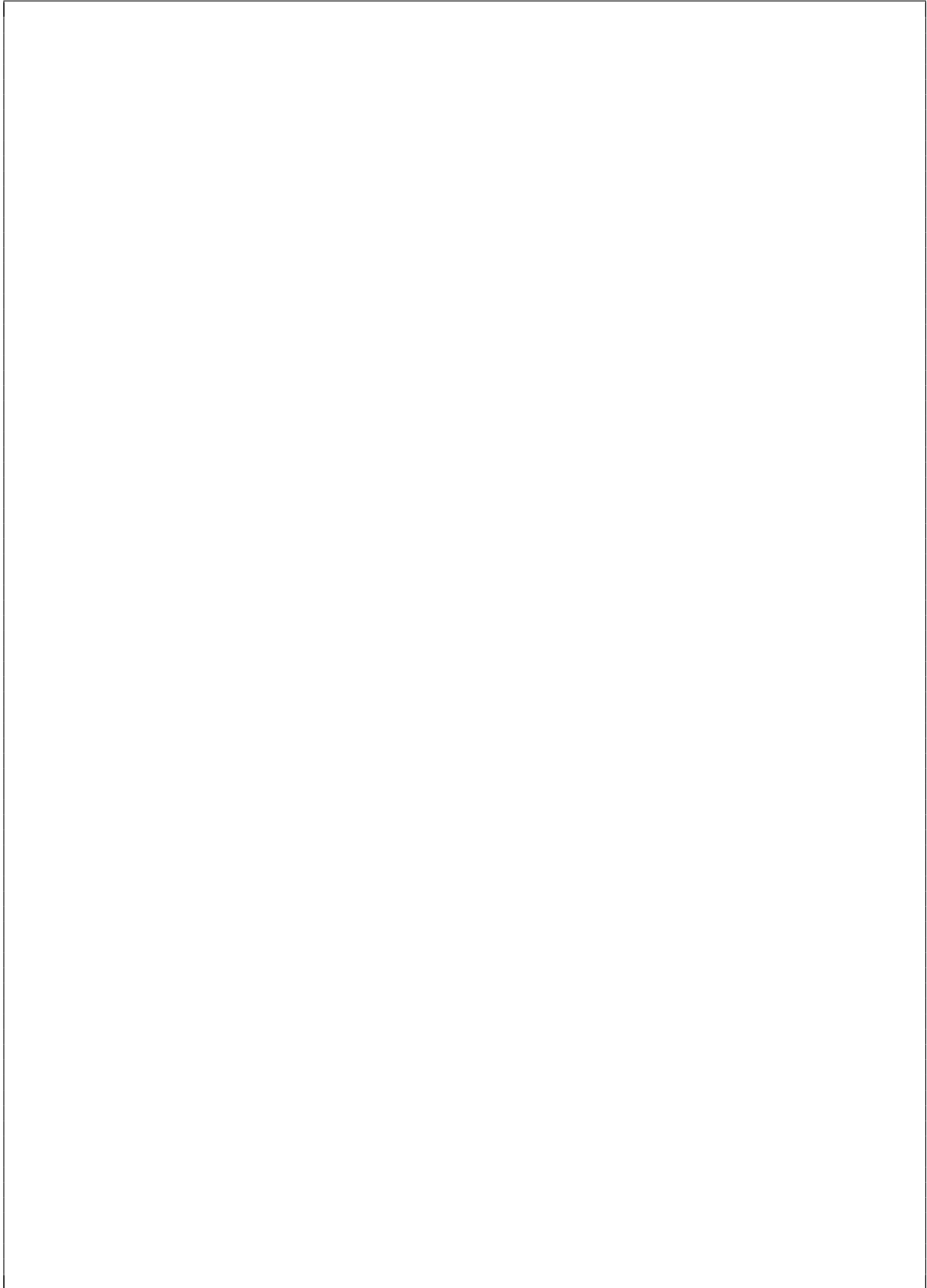
$$x_c(t) = \frac{\cos(40\pi t)}{160\pi t}$$

is sampled with a sampling period T to obtain the discrete-time signal

$$x[n] = \frac{\cos(\pi n/15)}{4\pi n/15}.$$

Determine a choice for T consistent with this information.

- d. Is your choice for T above unique? If so, explain why. If not, specify another choice of T consistent with the information given.



END OF PART A

Part B: Machine Learning

Empl ID: _____

Question 3 [10 marks]

Suppose you have an unknown sensor that is used to measure the distance to an object. The sensor produces voltages as outputs that you can measure. However, you do not know how to convert the voltages to distances. Figure 1. shows all of the collected training data and the results (A, B, C) of training a linear regression model three times using different training data randomly sampled from the full dataset.

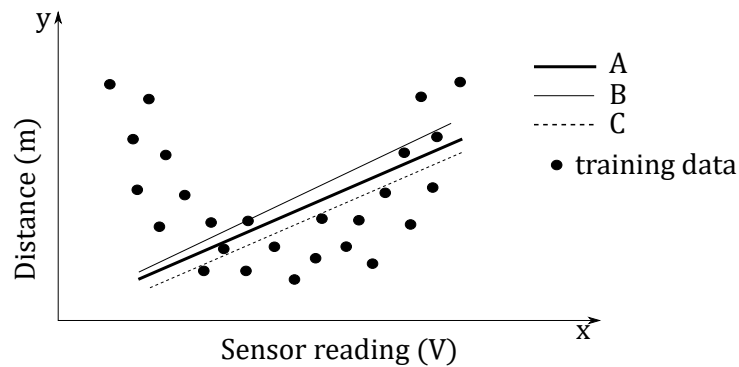


Figure 1: The output of a linear regression model trained three times with different samples of the training data

- a. Would you consider this model to have a high or low bias? Explain your answer. (2)

- b. Would you consider this model to have a high or low variance? Explain your answer. (2)

At this point, you decide to train a polynomial regression model with the model's output shown in Figure 2 when trained on three different splits of the training data.

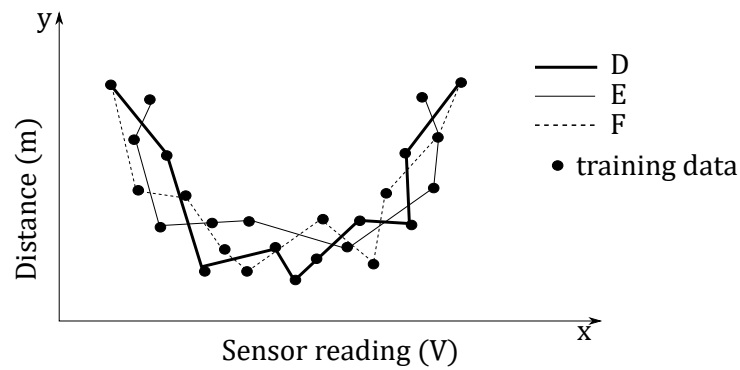


Figure 2: The output of a polynomial regression model trained three times with different samples of the training data

- c. Would you consider this model to have a high or low bias? Explain your answer. (2)

- d. Would you consider this model to have a high or low variance? Explain your answer. (2)

- e. Which of the models (linear or polynomial) would benefit the most from regularization? Explain your answer. (2)

Question 4 [10 marks]

Suppose you collect a data set for the previous distance measuring sensor, $\mathcal{D}$, consisting of input-output pairs (x_i, y_i) where x_i is the measurement from the sensor in, V, and y_i is the target variable, which in this case is the distance from the sensor to a wall in meters.

$$\mathcal{D} = \{(0.2, 3), (0.5, 2), (1.3, 1.3), (2, 4)\}$$

You then decide to fit a regression model of the form $h_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 x^2$ to it using gradient descent optimization methods. The model's parameters are randomly initialised such that

$$\theta_0 = 1, \theta_1 = 2, \theta_2 = 6$$

The objective function, J , you choose to optimize is given by the mean square error of the form:

$$J = \frac{1}{2N} \sum_{i=1}^N (h_{\theta}(x_i) - y_i)^2$$

Where N is the number of examples in a single training iteration.

Parameters are updated after **each** iteration, and a single iteration uses one example from the data set. The update rule is given by:

$$\theta_i := \theta_i - \eta \frac{\partial J}{\partial \theta_i}$$

Where $\eta = 0.1$. Answer the following questions

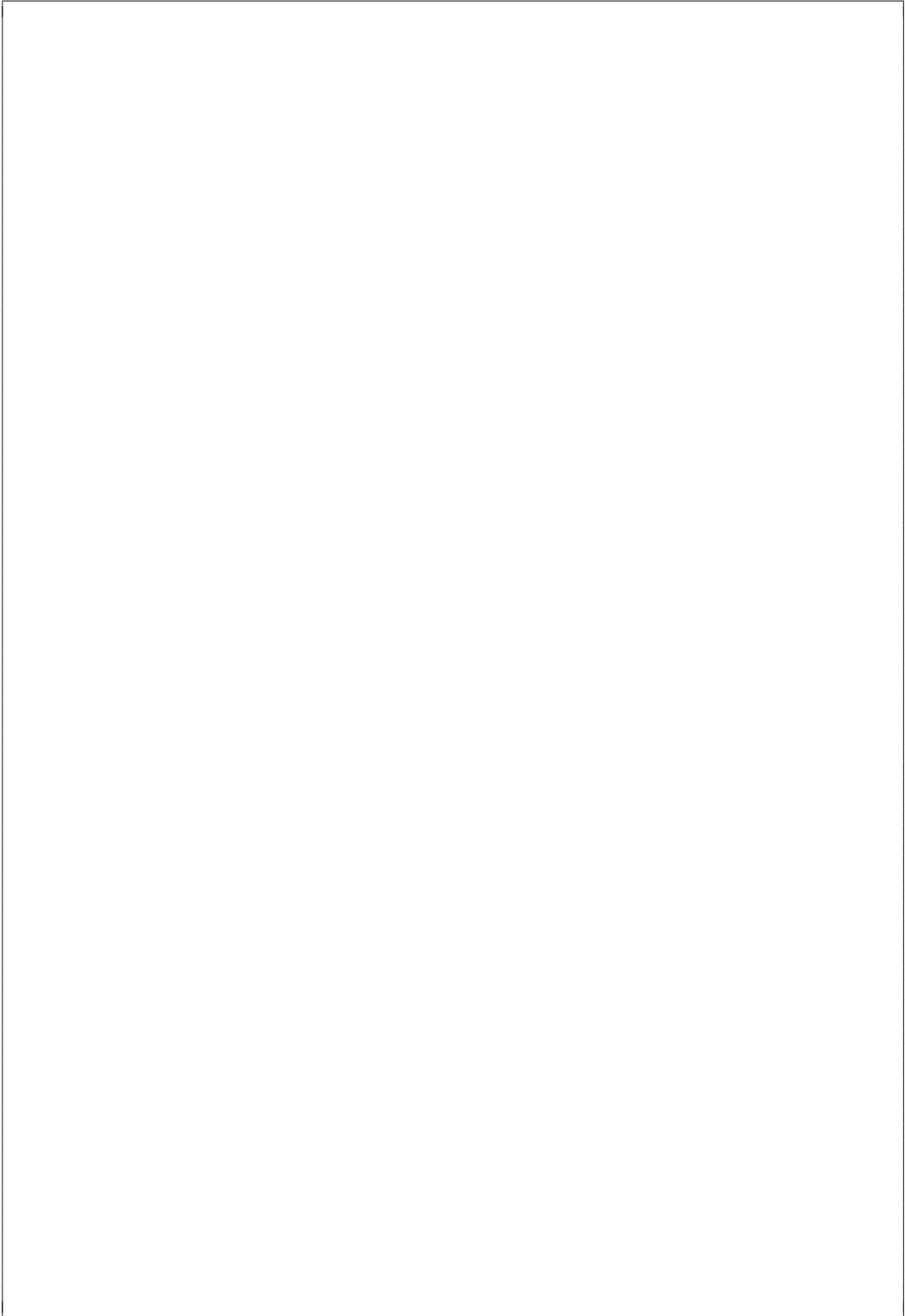
- a. Determine the expressions for the gradients of the objective function with respect to the model's parameters: $\frac{\partial J}{\partial \theta_0}$, $\frac{\partial J}{\partial \theta_1}$, and $\frac{\partial J}{\partial \theta_2}$. (3)

- b. Calculate the gradients of the parameters after one training iteration using the first training pair in the data set $(x_1, y_1) = (0.2, 3)$. Round your answers to two decimal places. (3)



- c. Calculate the values for θ_0 and θ_2 after the second training iteration where $(x_2, y_2) = (0.5, 2)$. You can assume the model output is given by $h_\theta(0.5) = 3.7$. (4)





END OF PART B

DSP information sheet

Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt \quad \Longleftrightarrow \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega \quad (\text{CTFT})$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega \quad (\text{DTFT})$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz \quad (\text{ZT})$$

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \quad \Longleftrightarrow \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with } W_N = e^{-j\frac{2\pi}{N}} \quad (\text{DFT})$$

Discrete-time Fourier transform properties

Sequences $x[n], y[n]$	Transforms $X(e^{j\omega}), Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d} X(e^{j\omega})$	Time shift
$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n+1)a^n u[n] \quad (a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n], y[n]$	Transforms $X(z), Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC = R_x	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC = $ z_0 R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC = $\frac{1}{R_x}$	Time reversal
$nx[n]$	$-z \frac{dX(z)}{dz}$	ROC = R_x	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC = R_x	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$ z < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$ z > 0$
$\cos(\omega_0 n) u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n) u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$ z > r $

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for } a \neq 1.$$